

Translated

From: e-maxx.ru

Maximum flow - Push-relabel algorithm

The push-relabel algorithm (or also known as preflow-push algorithm) is an algorithm for computing the maximum flow of a flow network. The exact definition of the problem that we want to solve can be found in the article [Maximum flow - Ford-Fulkerson and Edmonds-Karp](#).

In this article we will consider solving the problem by pushing a preflow through the network, which will run in $O(V^4)$, or more precisely in $O(V^2 E)$, time. The algorithm was designed by Andrew Goldberg and Robert Tarjan in 1985.

Definitions

During the algorithm we will have to handle a **preflow** - i.e. a function f that is similar to the flow function, but does not necessarily satisfies the flow conservation constraint. For it only the constraints

$$0 \leq f(e) \leq c(e)$$

and

$$\sum_{(v, u) \in E} f((v, u)) \geq \sum_{(u, v) \in E} f((u, v))$$

have to hold.

So it is possible for some vertex to receive more flow than it distributes. We say that this vertex has some excess flow, and define the amount of it with the **excess** function $x(u) = \sum_{(v, u) \in E} f((v, u)) - \sum_{(u, v) \in E} f((u, v))$.

In the same way as with the flow function, we can define the residual capacities and the residual graph with the preflow function.

The algorithm will start off with an initial preflow (some vertices having excess), and during the execution the preflow will be handled and modified. Giving away some details already, the algorithm will pick a vertex with excess, and push the excess to neighboring vertices. It will repeat this until all vertices, except the source and the sink, are free from excess. It is easy to see, that a preflow without excess is a valid flow. This makes the algorithm terminate with an actual flow.